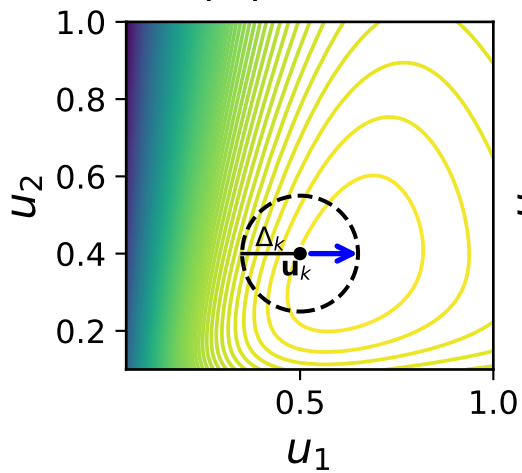
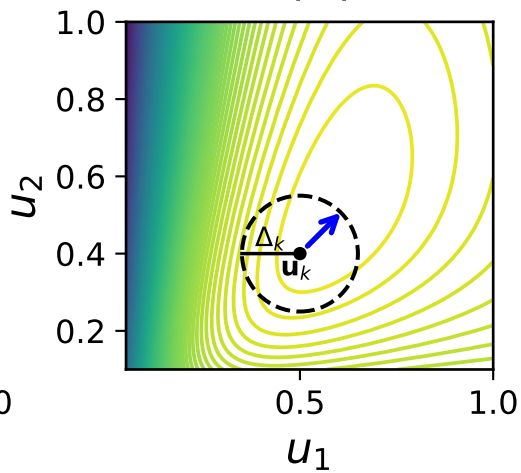
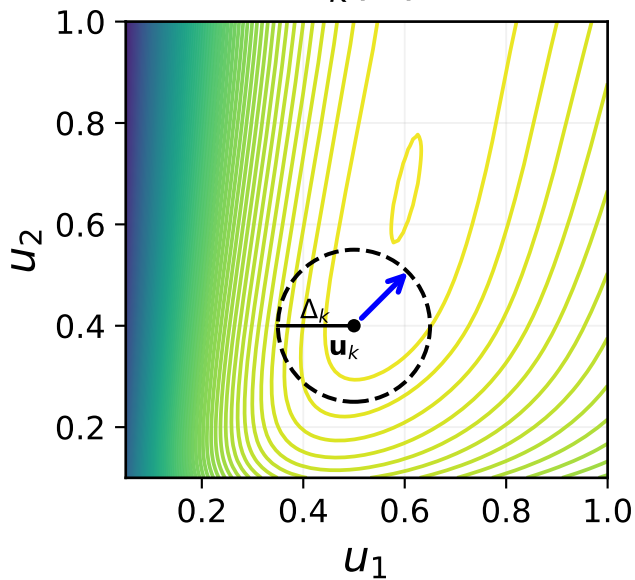


$\tilde{r}(\mathbf{u}) - PINN$  $s(\mathbf{u})$  $r_k(\mathbf{u})$ Oil production rate, $w_{o,out}$